

Breast Ultrasound Image Analysis:

Classification & Segmentation with Deep Learning Ensembles

A Two-Stage Deep Learning Approach · Group Project

Group Members

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Problem Statement, Motivation & Our Approach

Clinical Context

- Breast cancer is the most commonly diagnosed cancer among women worldwide. In 2022, approximately 2.3 million new cases and 670,000 deaths occurred, and new cases are projected to increase by 38% by 2050 [1].
- In the US, the 5-year relative survival rate is over 99% for localized breast cancer but falls to 32% for distant-stage disease [2].
- Ultrasound imaging is widely used — but manual interpretation is time-consuming & subjective.

Motivation

- Segmenting normal images is unnecessary and may increase false-positive predictions
- Single model can overfit on small datasets like BUSI (780 images)
- Solution: Classify first → segment only if abnormal
- Ensemble learning improves robustness & generalization

Our Approach

Stage 1 — Classification

- **Input:** breast ultrasound image
- **Output:** Normal / Benign / Malignant
- **Models:** EfficientNet-B4; Swin-Transformer Tiny; MobileViTv2-100

Stage 2 — Segmentation

- **Input:** benign / malignant image
- **Output:** pixel-level tumour mask
- **Models:** Attention U-Net; UNet++; TransUNet
- Skipped for normal class

Dataset Overview & EDA Findings

Dataset: BUSI (Breast Ultrasound Images) [3]

- Collected from 600 female patients, aged 25–75
- Data Download Source: MATLAB Embedded Dataset

780

Original images

3

Classes

256×256

Resized resolution

100%

Images have masks

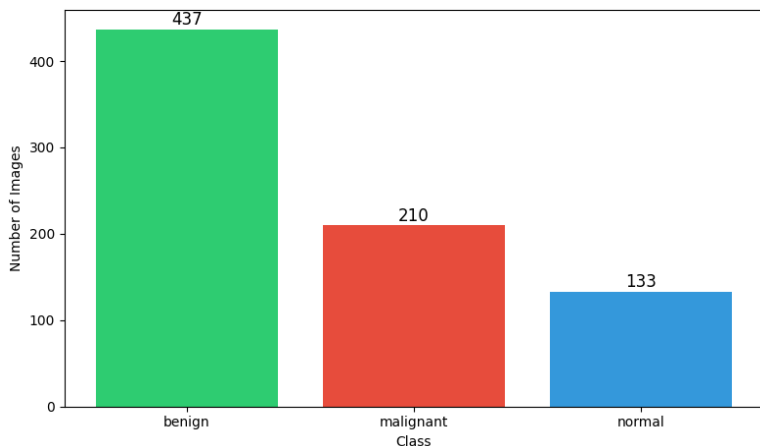


Figure 1. Class distribution in the original BUSI dataset

EDA Findings

- **Width:** 190–1048 px (mean 616)
- **Height:** 310–719 px (mean 501)
- **Multiple masks:** benign 16 cases, malignant 1 case (merged using OR operation)
- 1 duplicate image detected (Duplicate was kept to preserve the original dataset distribution.)
- 0 corrupt images

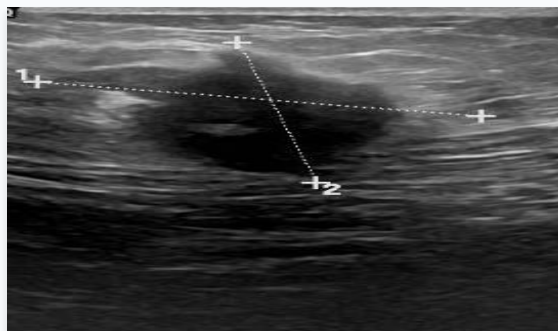
Sample Images & Masks

Each class has a distinct ultrasound appearance and corresponding segmentation mask.

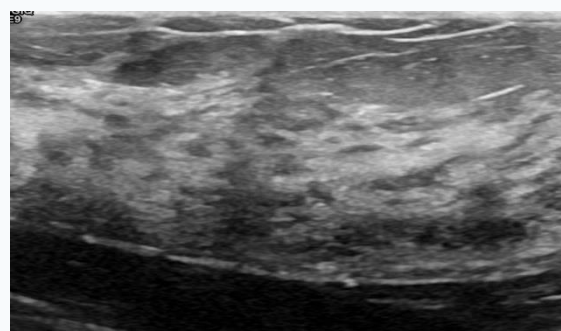
Benign



Malignant



Normal



Mask

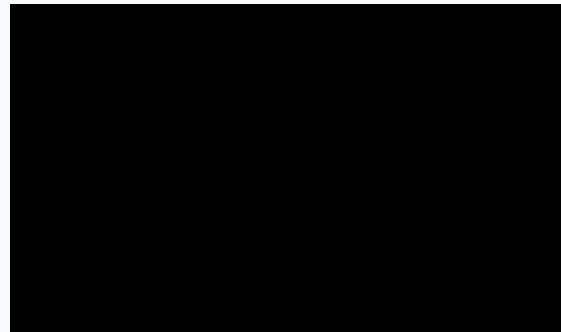
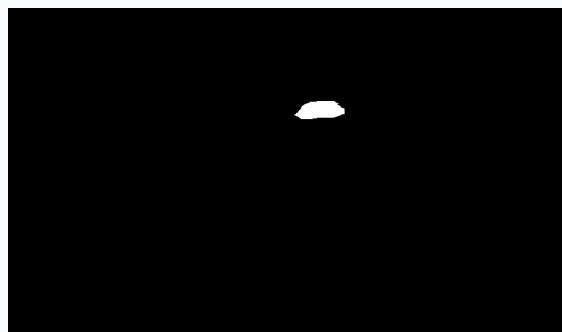


Figure 2. Representative ultrasound images and corresponding masks from the BUSI dataset

End to End DL Project Workflow

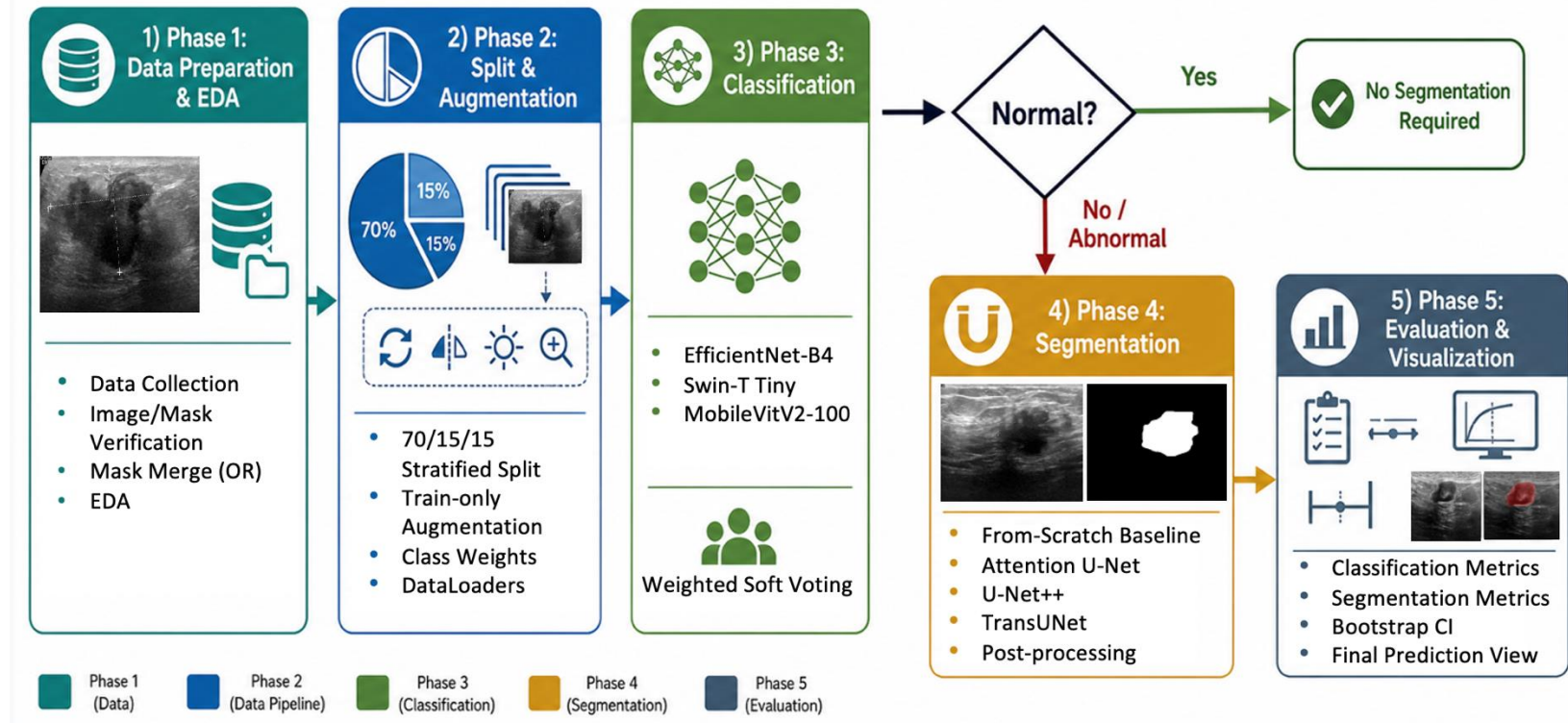
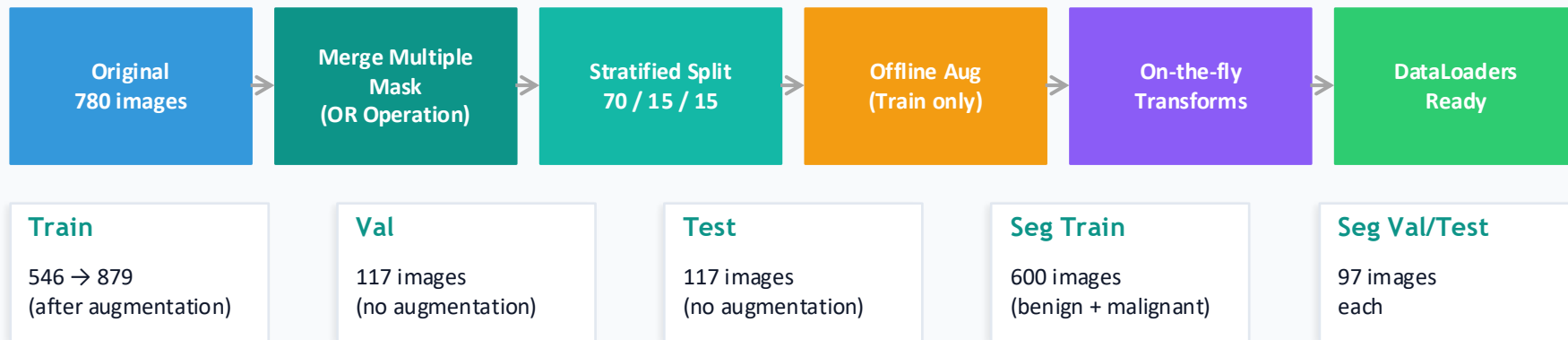


Figure 3. End-to-end deep learning project workflow

Data Pipeline: From Original Images to DataLoaders



Augmentation Strategy

Offline (new files, train only)

- Horizontal & Vertical Flip
- Rotation $\pm 15^\circ$
- Elastic Transform ($\alpha=50, \sigma=5$)
- Malignant: $\times 1 \rightarrow 147+147=294$
- Normal: $\times 2 \rightarrow 93+186=279$

On-the-fly (every epoch)

- GaussNoise (10–50 var)
- RandomBrightnessContrast ± 0.2
- GaussianBlur 3–5 kernel
- CLAHE clip=2.0
- ImageNet Normalization

Val / Test

- Resize 256×256 only
- ImageNet Normalization
- No geometric augmentation
- Ensures fair evaluation

Classification Models

EfficientNet-B4	Swin Transformer Tiny	MobileViTV2 - 100	Ensemble
CNN 17.554 M params Pretrained: ImageNet-1K Modified Head: Dropout(0.4) → Linear(3) Backbone: EfficientNet-B4 feature extractor	Transformer 27.52 M params Pretrained: ImageNet-1K Modified Head: Dropout(0.3) → Linear(3) Backbone: Swin-T feature extractor	CNN + ViT 4.39 M params Pretrained: ImageNet-1K Modified Head: Dropout(0.3) → Linear(3) Backbone: MobileViTV2-100 feature extractor	Weighted soft voting ensemble Input: Class probabilities from EfficientNet-B4, Swin-T, and MobileViT Weights: Based on validation accuracy Combination: Weighted average of softmax probabilities Final prediction: Class with the highest weighted-average probability Output classes: Benign, Malignant, Normal
Why this model? - Compound scaling — balances depth, width, resolution efficiently - Strong CNN baseline - Good accuracy-efficiency balance - Effective pretrained feature extractor	Why this model? - Shifted-window attention captures local structure while enabling cross-window context - Transformer-based comparison model - Captures spatial relationships with attention - Good for learning global image patterns	Why this model? - Lightweight hybrid model - Combines CNN local features + transformer context - Lower computational cost - Useful architecture diversity for ensemble	Why? improves robustness by combining, reduce single-model bias.

3-Phase Fine-tuning: Phase 1 — Freeze backbone (LR 1e-3, 10 ep) → Phase 2 — Partial unfreeze 30% (LR 1e-4, 10 ep) → Phase 3 — Full (LR 1e-5, 30 ep)
Loss: Weighted CrossEntropy | Optimizer: AdamW | Scheduler: CosineAnnealing

Classification – Training Results

86.32%

EfficientNet-B4

88.03%

Swin Transformer Tiny

90.60%

MobileViTv2-100

90.60%

Ensemble

Validation Loss

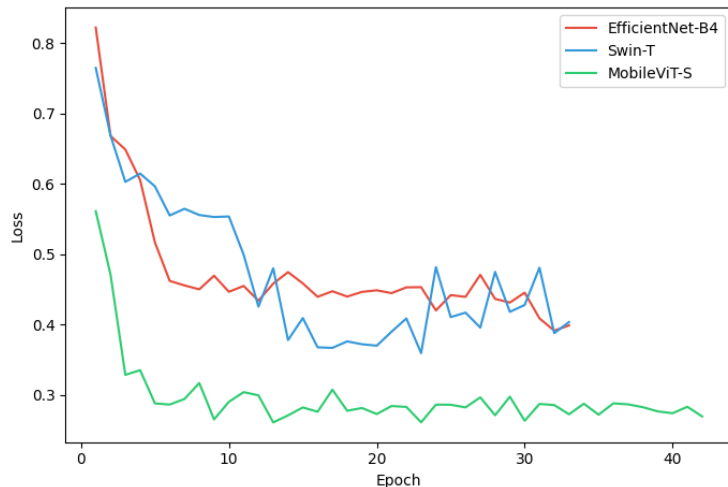


Figure 4. Validation-loss curves for the three classification models

Validation Accuracy

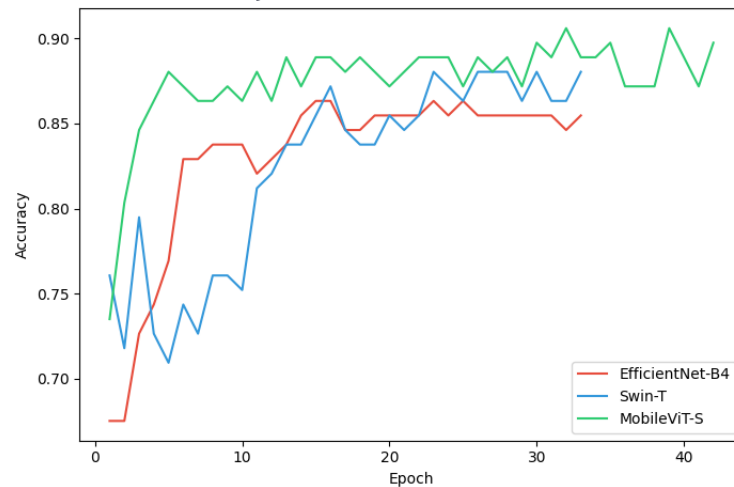


Figure 5. Validation-accuracy curves for the three classification models

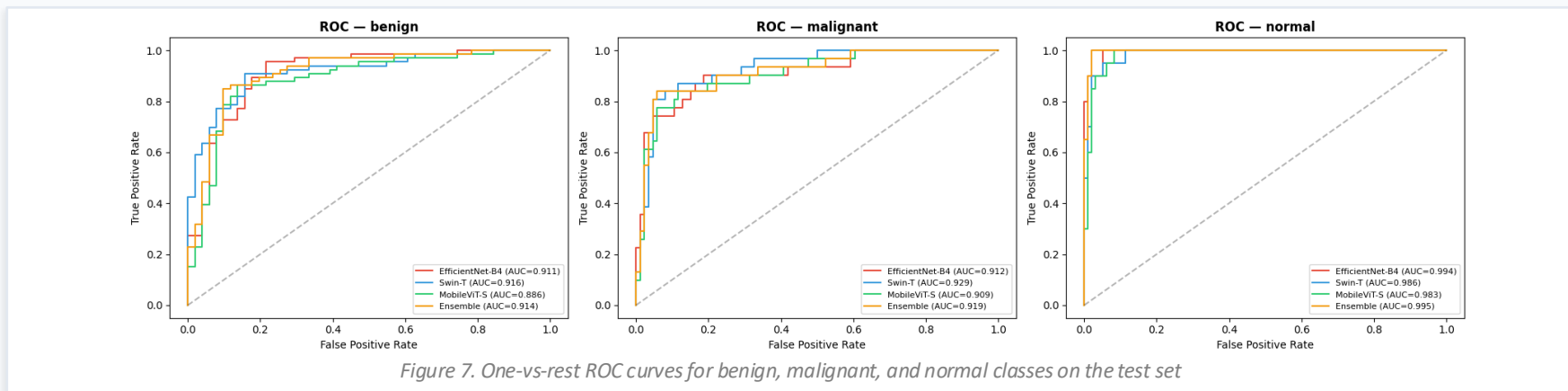
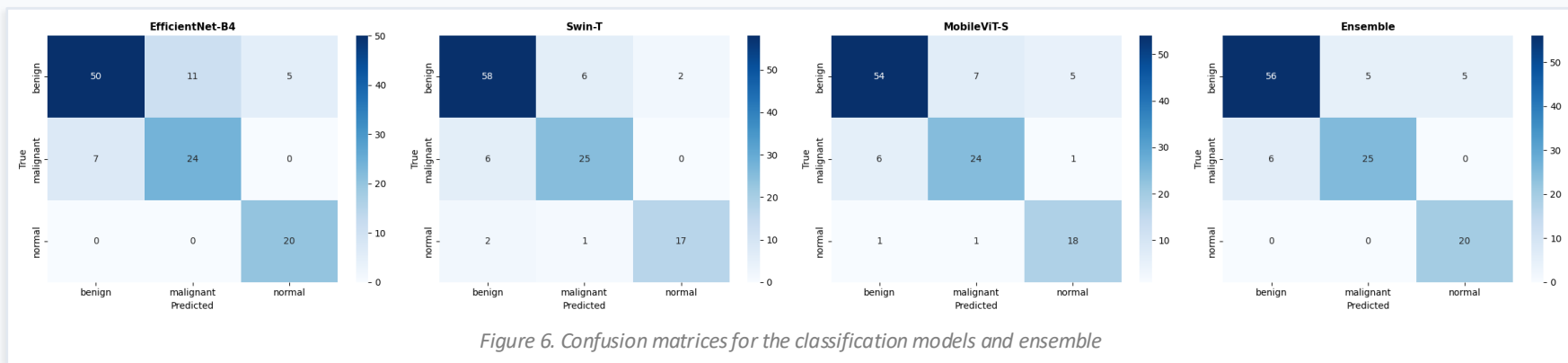
Observation: MobileViTv2-100 achieved the highest individual validation accuracy of 90.60%. The weighted ensemble also achieved 90.60%. EfficientNet-B4, Swin-T, and MobileViTv2-100 ran for 33, 33, and 42 epochs, respectively, before early stopping.

Classification Results – Test Set

Metric	EfficientNet-B4	Swin-T Tiny	MobileVitV2-100	Ensemble ✓
Accuracy	0.8034	0.8547	0.8205	0.8632
AUC-ROC (Macro)	0.9391	0.9438	0.9260	0.9425
Macro F1	0.8097	0.8481	0.8102	0.8612
MCC	0.6829	0.7503	0.7020	0.7729
Sensitivity — Benign	0.7576	0.8788	0.8182	0.8485
Sensitivity — Malig.	0.7742	0.8065	0.7742	0.8065
Sensitivity — Normal	1.0000	0.8500	0.9000	1.0000
Specificity — Benign	0.8627	0.8431	0.8627	0.8824
Specificity — Malig.	0.8721	0.9186	0.9070	0.9419
Specificity — Normal	0.9485	0.9794	0.9381	0.9485

Table 1. Test-set performance of the classification models and weighted ensemble

Classification – Confusion Matrices & ROC Curves



Segmentation Models

Architecture of Attention U-Net (From Scratch) - 24.525M params

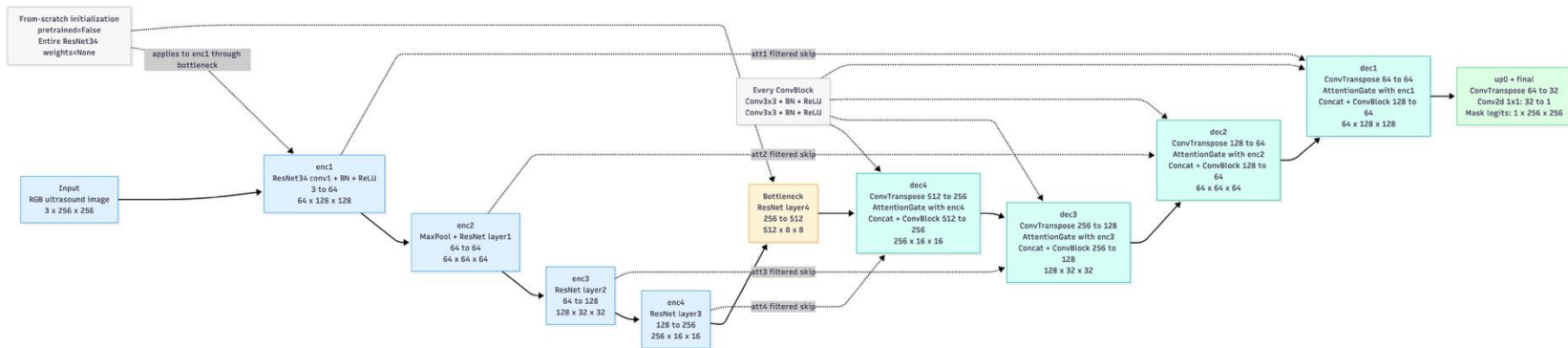


Figure 8. Architecture of the from-scratch Attention U-Net

Type: CNN-based encoder-decoder segmentation model

Purpose: Controlled comparison for evaluating transfer learning

Encoder: Randomly initialized ResNet34 encoder

Weights: Initialization: From scratch (pretrained=False, weights=None)

Attention: Attention-gated skip connections

Decoder: ConvTranspose + ConvBlock

Output: 1-channel tumour mask

Loss: BCE + Dice Loss

Optimizer: AdamW, LR = 1e-3, weight decay = 1e-2

Scheduler: CosineAnnealingLR

Training: Single-phase training, max 50 epochs

Segmentation Models (Cont.)

Architecture of Attention U-Net - 24.525M params

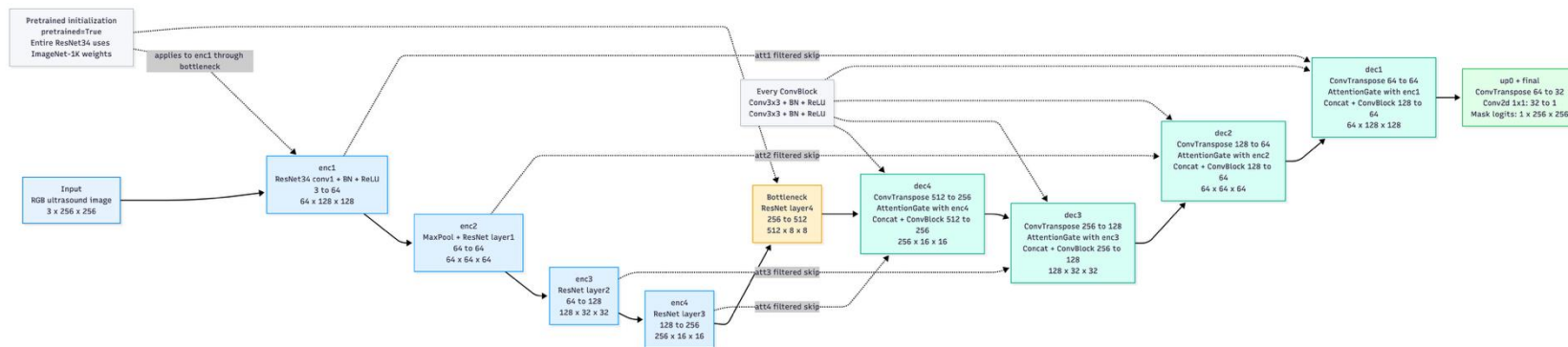


Figure 9. Architecture of the pretrained Attention U-Net

Type: CNN-based encoder-decoder segmentation model

Purpose: Tumour region segmentation

Encoder: ResNet34 encoder and bottleneck

Weights: ImageNet-1K V1 (pretrained=True)

Attention: Attention-gated skip connections

Decoder: Transposed convolution + ConvBlock

Output: 1-channel tumour mask

Loss: 0.5 BCE + 0.5 Dice loss

Optimizer: AdamW, weight decay 1e-2

LR: 1e-3 → 1e-4 → 1e-5

Scheduler: CosineAnnealingLR

Training: 3 phase fine-tuning (freeze encoder, partial unfreeze, full fine tune)

Segmentation Models (Cont.)

Architecture of U-Net++ (26.156M params)

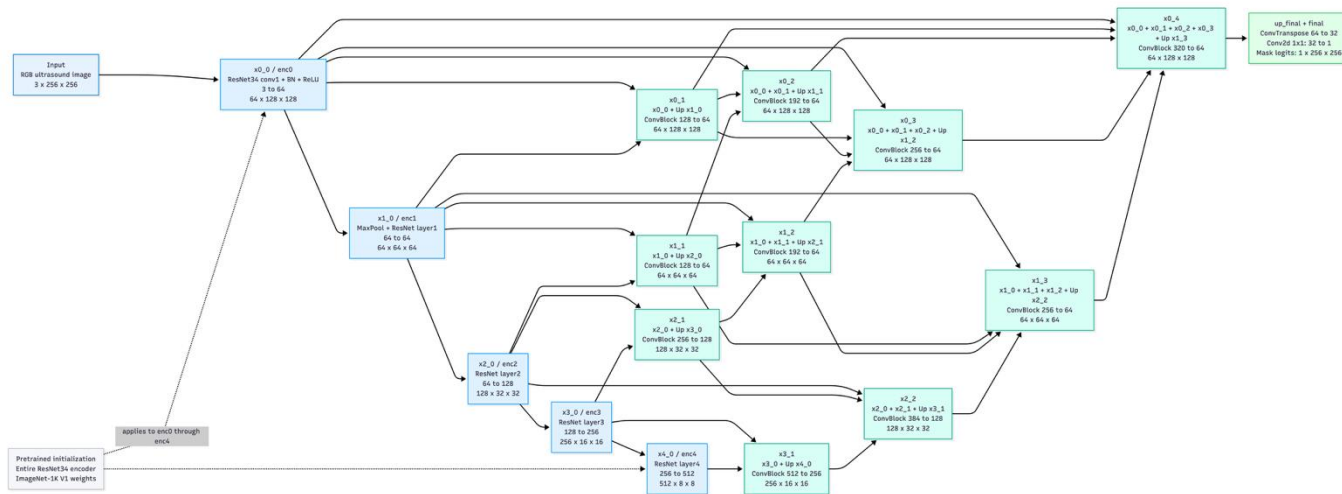


Figure 10. Architecture of the pretrained U-Net++

Type: CNN-based nested encoder-decoder segmentation model

Purpose: Tumour region segmentation

Encoder: ResNet34 encoder pretrained on ImageNet-1K V1

Weights: ResNet34 pretrained weights

Architecture style: Nested decoder with dense multi-scale skip connections

Decoder: Nested decoder with multiple skip pathways

Output: 1-channel tumour mask

Loss: 0.5 BCE + 0.5 Dice loss

Optimizer: AdamW, weight decay 1e-2

Training: 3 phase fine-tuning (freeze encoder, partial unfreeze, full fine tune)

Segmentation Models (Cont.)

Architecture of Trans U-Net (67.8M params)

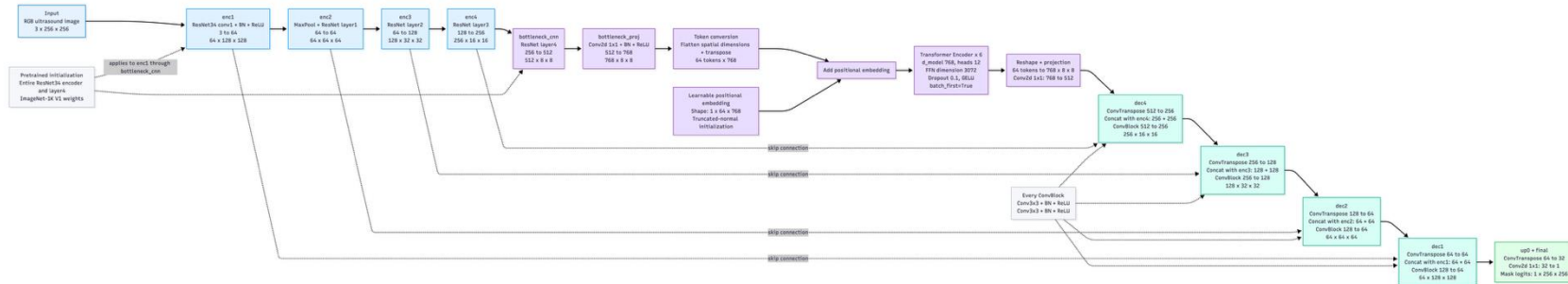


Figure 11. Architecture of TransUNet with a pretrained ResNet34 encoder

Type: CNN + Transformer hybrid segmentation model

Purpose: Tumour region segmentation

Encoder: ImageNet-1K V1 pretrained ResNet34 encoder and CNN bottleneck

Weights: ResNet34 pretrained weights

Architecture style: CNN-Transformer hybrid U-Net architecture

Decoder: Transposed convolution + ConvBlock

Output: 1-channel tumour mask

Loss: 0.5 BCE + 0.5 Dice loss

Optimizer: AdamW, weight decay $1e-2$

Training: 3 phase fine-tuning (freeze encoder, partial unfreeze, full fine tune)

Segmentation – Training Results

Dice 0.7618

Attention U-Net · 24 epochs

Dice 0.7747

UNet++ · 49 epochs

Dice 0.7705

TransUNet · 34 epochs

Dice 0.7319

From-Scratch · 41 epochs

Validation Loss

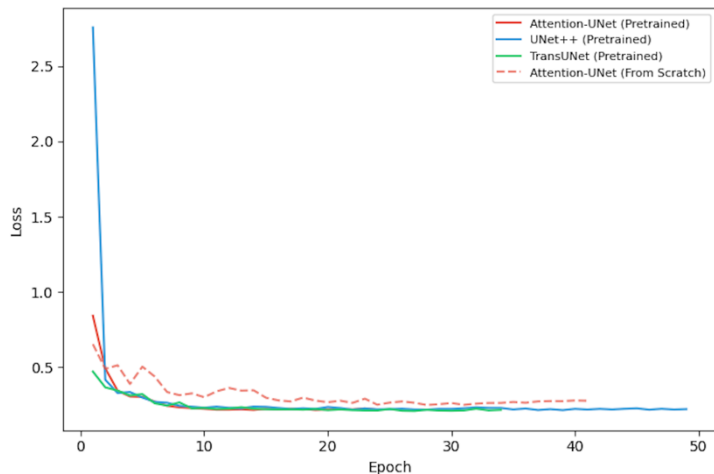


Figure 12. Validation-loss curves for the four segmentation experiments

Validation Dice Score

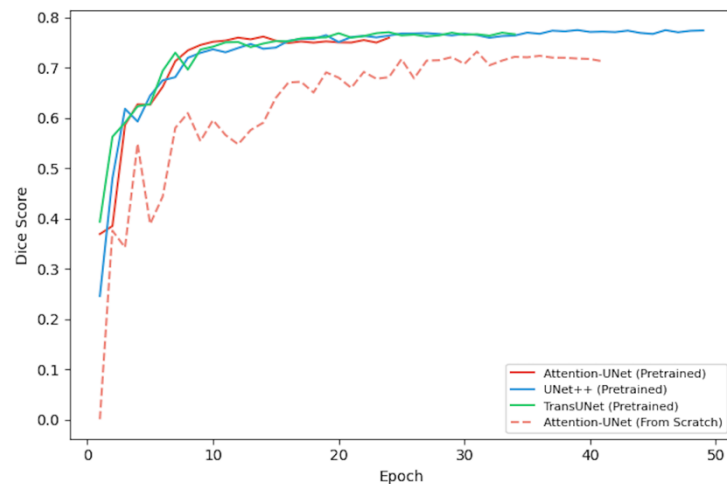


Figure 13. Validation-Dice curves for the four segmentation experiments

Observation: U-Net++ achieved the highest validation Dice (0.7747), closely followed by TransUNet (0.7705). The pretrained Attention U-Net outperformed its from-scratch counterpart by 0.0299 Dice.

Segmentation Results – Test Set

Metric	Att-UNet	UNet++ ✓	TransUNet	Ensemble	Scratch
Dice	0.7602	0.7933	0.7788	0.7770	0.7459
IoU	0.6696	0.7035	0.6893	0.6911	0.6571
Pixel Accuracy	0.9595	0.9644	0.9627	0.9631	0.9537
Sensitivity	0.7859	0.8128	0.7718	0.7837	0.7726
Specificity	0.9816	0.9838	0.9871	0.9852	0.9799
Hausdorff95	28.64	25.00	26.02	26.82	26.60

Table 2. Test-set performance of the segmentation models, ensemble, and scratch baseline

Best overall model by Dice: U-Net++ – Dice 0.7933, IoU 0.7035, HD95 25.00 px

Segmentation — Visual Results (Val. set samples)

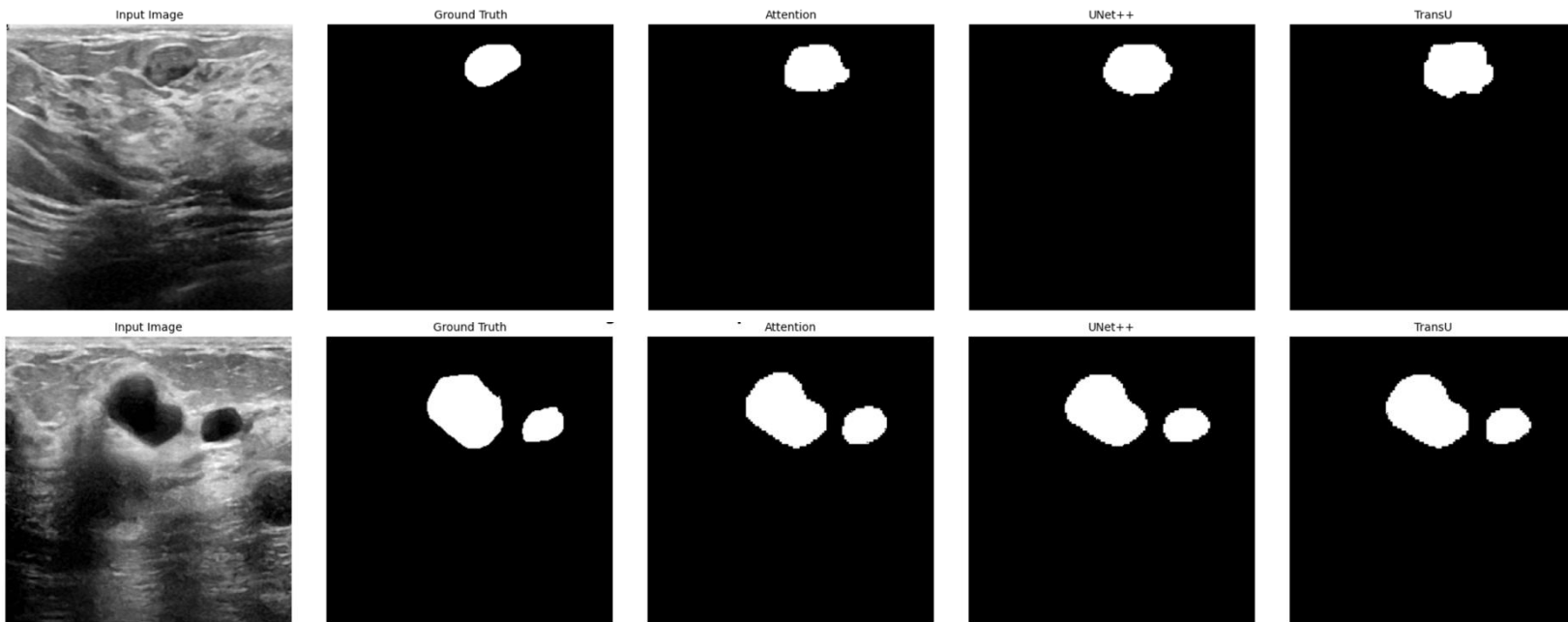


Figure 14. Qualitative segmentation results from the individual models

Segmentation – Test-Set Metric Comparison

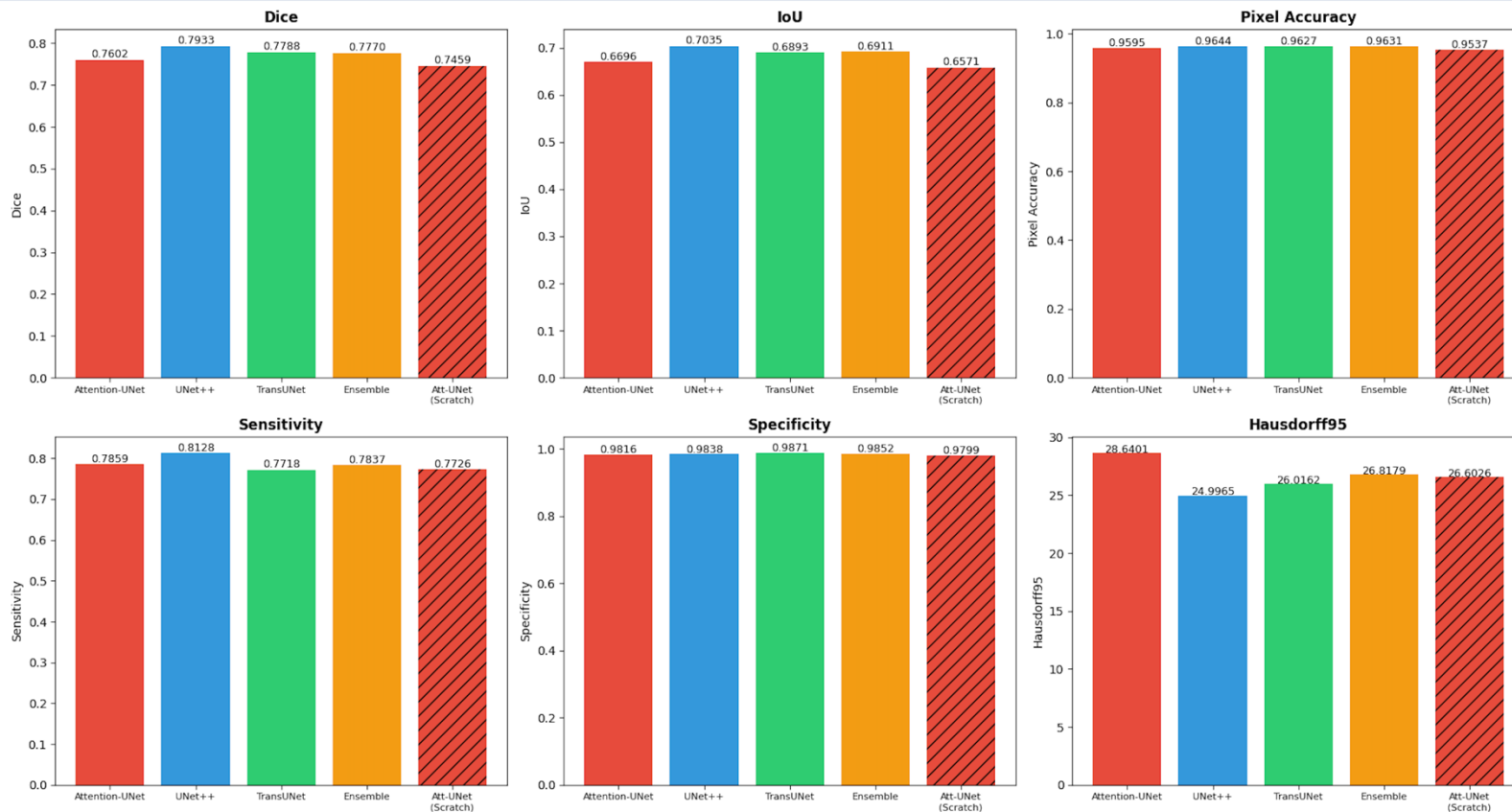


Figure 15. Test-set comparison of segmentation performance across individual models, ensemble, and scratch baseline

Full Pipeline — End-to-End Demo

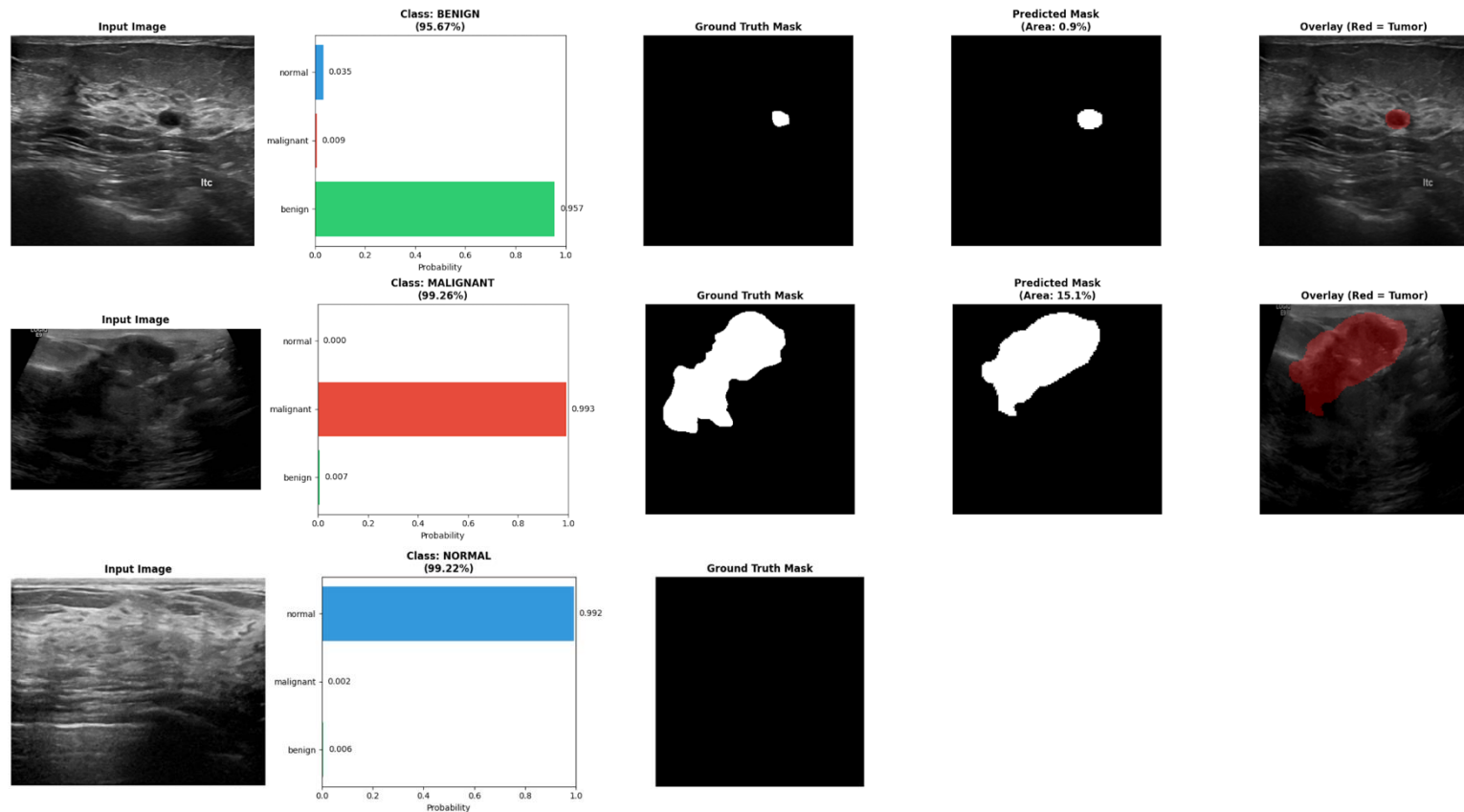


Figure 16. End-to-end pipeline predictions for benign, malignant, and normal samples

Alternative Experiments – From-Scratch vs Pretrained

We trained Attention-UNet without any pretrained weights to quantify the value of transfer learning on our small dataset.

Metric	Attention-UNet (From Scratch)	Attention-UNet (Pretrained)	Δ Improvement
Dice	0.7459	0.7602	+0.0143
IoU	0.6571	0.6696	+0.0125
Pixel Accuracy	0.9537	0.9595	+0.0058
Sensitivity	0.7726	0.7859	+0.0133
Specificity	0.9799	0.9816	+0.0017
Hausdorff95	26.60	28.64	+2.04 px (Scratch lower/better)
Val Dice (best)	0.7319	0.7618	+0.0299
Total epochs run	41	24	17 fewer (faster)

Table 3. Pretrained versus from-scratch Attention U-Net performance

Analysis

- Limited original abnormal training data: 453 images, expanded to 600 entries after augmentation
- The ImageNet-pretrained encoder provides transferable edge, texture, and shape features
- The pretrained model used three-phase fine-tuning, while the scratch model used single-phase full training
- The scratch model achieved lower HD95 on this test set; the reason requires further analysis.
- **Conclusion:** Pretraining improved most overlap-based metrics, although the scratch model achieved better HD95

Discussion – Results & Limitations

✓ Positive Results

- Strong classification accuracy
 - Classification ensemble achieved 86.32% test accuracy, 0.8612 Macro F1, and 0.7729 MCC.
- 100% normal recall
 - All 20 normal test images were correctly identified, giving normal-class sensitivity of 1.000.
- Robust segmentation
 - U-Net++ achieved the best overall segmentation performance: Dice 0.7933, IoU 0.7035, and HD95 25.00 px.
- Transfer learning benefit observed
 - Pretrained Attention U-Net improved Dice by 0.0143 and IoU by 0.0125 over the scratch version; however, the scratch model achieved lower HD95.
- Classification ensemble provided a modest accuracy gain
 - The ensemble improved test accuracy from the best individual result of 85.47% to 86.32%, a gain of 0.85 percentage points.

✗ Negative Results & Limitations

- Benign/malignant confusion
 - The ensemble misclassified 5 benign images as malignant and 6 malignant images as benign, indicating overlap between abnormal classes.
- Limited classification-ensemble gain
 - The ensemble and MobileViTv2-100 both achieved 90.60% validation accuracy; on the test set, the ensemble improved only 0.85 percentage points over Swin-T.
- Small dataset
 - The dataset contains only 780 original images, with 117 classification-test images and 97 segmentation-test images, limiting generalizability.
- Pipeline error propagation
 - A false-normal prediction would skip a required segmentation, while a false-abnormal prediction would trigger unnecessary segmentation.

Conclusion

86.32%

Classification Accuracy
(Ensemble)

0.9438

AUC-ROC (Macro)
(Swin-T)

0.7933

Best Seg Dice
(UNet++)

7

Training Experiments
(3 CLF + 4 SEG)

What we built

- A two-stage breast-ultrasound pipeline: three-class classification followed by conditional lesion segmentation.
- Stage 1 combined CNN, Transformer, and hybrid classifiers using validation-accuracy-weighted soft voting.
- Stage 2 combined three pretrained segmentation models using equal probability-mask averaging and post-processing.
- Pretrained Attention U-Net improved test Dice by 0.0143 over the scratch experiment, although scratch achieved better HD95.
- U-Net++ outperformed the segmentation ensemble, while the classification ensemble provided only a modest accuracy gain.

Future Work:

- External validation
- K-fold evaluation
- Threshold and ensemble-weight optimization
- Semi-supervised learning
- Prospective clinical evaluation
- Multi-lesion-aware post-processing

References

- [1] J. Kim, A. Harper, V. McCormack, H. Sung, N. Houssami, E. Morgan, M. Mutebi, G. Garvey, I. Soerjomataram, and M. M. Fidler-Benaoudia, "Global patterns and trends in breast cancer incidence and mortality across 185 countries," *Nature Medicine*, vol. 31, pp. 1154–1162, 2025, doi: 10.1038/s41591-025-03502-3.
- [2] American Cancer Society, *Breast Cancer Facts & Figures 2024–2025*. Atlanta, GA, USA: American Cancer Society, 2024.
- [3] W. Al-Dhabyani, M. Gomaa, H. Khaled, and A. Fahmy, "Dataset of breast ultrasound images," *Data in Brief*, vol. 28, Art. no. 104863, Feb. 2020, doi: 10.1016/j.dib.2019.104863.

Thank You

Questions & Discussion

Dataset: BUSI (Breast Ultrasound Images) | Two-Stage Deep Learning Pipeline

CLF Ensemble Test Accuracy

86.32%

CLF Ensemble Macro AUC

0.9425

Best Test Dice (U-Net++)

0.7933

Model Training Experiments

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